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Analysing data sources' suitability to support forest policy decision-making in the Czech Republic

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SUMMARY

Growing stock, increment, realised felling and potential wood supply are important indicators for strategic policy decision-making in wood-based industry sectors. The aim of the article is to analyse two different data sources, the Forest Management Plans and Guidelines (FMPG) and the National Forest Inventory (NFI), used for the evaluation and prediction of these indicators in the Czech Republic. The differences between the two sources were analysed in terms of their credibility and suitability to support policy decision-making at the state, regional and local levels. Although the FMPG play an important role in policy decision-making, they are not able to determine indicators such as damage inflicted by game, forest regeneration or changes in soil conditions. The NFI, on the other hand, should provide up-to-date results every five years. Unlike the FMPG, where the collected information is reduced, the NFI is a flexible tool for the incorporation of new requirements in measurements.

Keywords: forest category, forest management plans and guidelines, forestry dynamic model, growing stock, National Forest Inventory

Analyser la capacité des sources de données à soutenir les prises de décision de la gestion forestière dans la République Tchèque

M. SYNEK et M. HRIB

La croissance des stocks, l'incrément, la coupe réalisée et la disponibilité potentielle de bois sont des indicateurs importants dans les prises de décisions de gestion stratégique dans les secteurs industriels basés sur le bois. Le but de cet article est d'analyser deux différentes sources de données, les Plans et lignes de gestion forestière (FMPG) et l'Inventaire national des forêts (NFI), utilisés pour l'évaluation et la prédiction de ces indicateurs en République Tchèque. Les différences entre les deux sources ont été analysées en termes de leur crédibilité et de leur capacité à soutenir les prises de décisions politiques aux niveaux local, régional et d'état. Bien que les FMPG jouent un rôle important dans les prises de décision politiques, elles ne sont pas à même de déterminer les indicateurs tels que le dommage infligé par le gibier, la régénération forestière, ou les changements de condition du sol. Le NFI devrait par contre fournir des résultats bien à jour tous les cinq ans. Au contraire des FMPG, dans lesquelles l'information recueillie est réduite, le NFI est un outil flexible, permettant l'incorporation de nouvelles exigences de mesures.

Análisis de la idoneidad de las fuentes de datos para apoyar la toma de decisiones en materia de política forestal en la República Checa

M. SYNEK y M. HRIB

Las existencias en crecimiento, el incremento, las talas realizadas y el potencial de suministro de madera son indicadores importantes para la toma de decisiones políticas estratégicas en los sectores de la industria maderera. El objetivo de este artículo es analizar dos fuentes de datos diferentes, a saber, los Planes y Directrices de Gestión Forestal (FMPG, por sus siglas en inglés) y el Inventario Forestal Nacional (IFN), utilizados para la evaluación y predicción de estos indicadores en la República Checa. Las diferencias entre las dos fuentes se analizaron en términos de su credibilidad e idoneidad para apoyar la toma de decisiones políticas a nivel nacional, regional y local. Aunque los FMPG desempeñan un papel importante en la toma de decisiones políticas, no son capaces de determinar indicadores como los daños causados por la caza, la regeneración de los bosques o los cambios en las condiciones del suelo. El IFN, por otra parte, debe proporcionar resultados actualizados cada cinco años. A diferencia de los FMPG, para los que la información recogida es limitada, el IFN es una herramienta flexible en cuanto a la incorporación de nuevos requisitos en las mediciones.

INTRODUCTION

Growing stock, increment, realised felling and potential wood supply are evaluated and modelled in many countries using different national models and input data. Growing stock, increment and realised felling are also regularly used as the main indicators of sustainable forest management in international reports, such as the State of Europe's Forests (SoEF) or Forest Resources Assessment (FRA). The Forest Europe has published a report entitled State of Europe's Forests since 2003, where every four years a current status and changes in European forests are evaluated by using the set of qualitative and quantitative indicators. The last report State of Europe's Forests was published in 2015. The Food and Agriculture Organization of the United Nations (FAO) has been monitoring the world's forests at 5 to 10 year intervals since 1946. Since 1990 the report Global Forest Resources Assessments has been produced every five years in order to describe the world's forests and their changes in a consistent way. The last report was also published in 2015 and for the first time data for both the reports (SoEF and FRA) were collected in mutually beneficial way.

On the basis of the national reports collected for the last publication of State of Europe's Forests 2015 it can be evaluated which data sources have been used for international reporting on growing stock, increment, fellings and removals. The data were collected for the year 2012 and extrapolated following the national trends to 2015.

In SoEF a growing stock is defined as "Volume over bark of all living trees with a minimum diameter of 10 cm at breast height (or above buttress if these are higher), including the stem from ground level up to a top diameter of 0 cm, excluding branches". The main source of information used by different countries to estimate the growing stock are National Forest Inventories (NFIs), which were used by twenty one countries and as additional sources by other four countries. Stand inventories were used by eight countries and remaining countries used other sources of information to estimate the growing stock. Kuliešis *et al.* (2016) stated that data obtained from stand inventories usually underestimated growing stock by 15–20%. Several NFIs also collect data for trees below the Diameter Breast Height (DBH) threshold, usually for a regeneration survey (Gschwantner *et al.* 2016). These data can be used for the estimation of stem volume (Tomter *et al.* 2012) or increment estimation trees below DBH threshold by applying different model types.

In SoEF a gross annual increment (GAI) is defined as "Average annual volume of increment over the reference period of all trees measured to minimum diameters as defined for Growing stock" and natural losses as "Average annual losses to the growing stock during the given reference period, measured to minimum diameters as defined for Growing stock, due to mortality from causes other than cutting by man, e.g. natural mortality, diseases, insects attacks, fire, wind throw or other physical damage". The main sources of information used by most countries for the GAI estimation are also NFIs, which were used by nineteen countries. Four countries have calculated the GAI as a percentage share of the growing

stock and other eight countries which were able to report on increment have used other sources of information, e.g. yield tables or tree rings measuring. Kuliešis *et al.* (2016) mentioned that data on GAI including its components, estimated from stand inventories, have low reliability and are characterized by frequent systematic errors compared to NFIs. According to Tomter *et al.* (2016) the most reliable system of information on the various components of annual increment are NFIs based on permanent plots. Thus all conclusions about forest growth and its use should be based on this information.

SoEF defines fellings as "Average standing volume of all trees, living or dead, measured over bark to minimum diameters as defined for "Growing stock" that are felled during the given reference period, including the volume of trees or parts of trees that are not removed from the forest, other wooded land or other felling site, including silvicultural and pre-commercial thinnings and cleanings left in the forest, and natural losses that are recovered (harvested)". According to this definition, only NFI's repeated measurements can fulfil all required fellings components. In SoEF 2015 ten countries reported volumes of fellings based on NFIs, twenty five countries based on other sources of information, usually national removal statistics and seven countries not reported the figures on fellings at all. Tomter *et al.* (2016) analysed the source of information on fellings in SoEF 2011 and obtained quite similar results. Seven countries used NFIs while removal statistics was used by nineteen countries and four countries used other various methods.

According to SoEF "the term "removal" differs from "felling" as it excludes harvesting losses (stemwood) and trees that were felled but not removed and it includes removals from fellings in earlier periods and from trees dead due to or damaged by natural causes". Compared to fellings, removals are reported under bark. Removals are reported every year as basic part of forest products statistics excluding NFIs as suitable sources of information on removals. The disadvantages of using NFI data are that the estimates will describe an average for a defined period between repeated measurements. Thus, annual estimates cannot be provided from the NFI data and regional estimates with a reasonable degree of reliability are also not possible (Tomter *et al.* 2016).

Potential wood supply is modelled on the national level using different models, which can be divided into three groups.

The first group of models consists of forest dynamic models which include growth and yield models and are designed to simulate stand development under alternative managements and growth conditions. This category includes the models such as SILVA in Germany (Pretzsch *et al.* 2002), PrognAus in Austria (Ledermann 2006), SIBYLA in Slovakia (Fabrika 2003, Fabrika and Adamec 2015), the MOTTI-stand simulator in Finland (Salminen *et al.* 2005) or several models developed under the CASIS platform in France (Dufour-Kowalski *et al.* 2012), e.g. OptimiSt.

The second group contains models which integrate the combination of stand and forestry dynamics models, more or less, such as Heureka in Sweden (Wikström *et al.* 2011)

which combines an interactive stand simulator and a simulator for regional analyses, or MELA in Finland (Redsven *et al.* 2012) which is used for calculating the timber production analysis based on the data of national forest inventories.

The third group includes matrix models of forestry dynamics using a transition matrix which expresses the probability of a unit leaving its current position within the matrix to join a different position and, thus, acquiring the characteristics of this new category. Individual trees, stands or other forest areas, e.g. age-volume classes, can be used as a basic unit. The European Forestry Dynamic Model (EFDM) (Packalen *et al.* 2014), EFISCEN (Schelhaas *et al.* 2007) or the Swedish matrix growth model (Sallnäs 1990) belongs to this category.

All of the above mentioned groups of models are used separately for the evaluation and prediction of forest stand indicators or as a part of a more complex decision support system (DSS). DSS enables one to model different scenarios of future development (e.g. climate change) or can be designed as equilibrium models, where the potential wood supply is compared with the results of the potential wood demand models.

In the Czech Republic, a potential wood supply was drawn up several times in the past for various prediction times. One of the first publications dealing with the topic was the Forest Inventory, issued in 1950. This inventory was based on forest management plans and included all forests in Bohemia and Moravia with a size above 10 ha (Lesprojekt, 1958). In 1960, a follow-up inventory was performed for planning forest management, based again on forest management plans. In addition to the forest inventories, comprehensive forest management plans based on new legislation were first introduced in 1978, and further prepared in five-year cycles. An integral part of the comprehensive forest management plans was the information on logging volumes for the next ten years and the potential wood supply for the following decades (Green Report 1994, Nymburský 1983).

The next study, with the prediction of the potential wood supply in the Czech Republic, was made in 2006 (Vašíček *et al.* 2006). In this study, the development of logging perspectives until 2065 was modelled using the matrix model and following the database of Forest Management Plans and Guidelines (FMPG). The guidelines are elaborated upon as a simplified form of forest management plans for small-scale forest owners with a forest area under 50 ha. Tests were carried out on the effect of a reduced rotation period in the selected management sets, an increased volume of tending measures up to 40 years, the combination of these two factors, and the use of the growing stock from the first cycle of the National Forest Inventory (NFI1).

A more detailed study was published in 2014 (Synek *et al.* 2014). The potential wood supply was not only derived as the total allowable cut for the next two decades, but also separately for coniferous and broadleaved tree species, for individual main tree species as well as the main assortments (roundwood, pulpwood, fuelwood). In this study, the data from the forest management plans and guidelines were used together with an advanced matrix model.

Adolt *et al.* (2014) compared the results from the study (Synek *et al.* 2014) with the preliminary results of the second cycle of the National Forest Inventory (NFI2). The results from the plots surveyed in 2013 were used for comparison, representing only a quarter of the final number of NFI2 plots. EFDM was used for evaluation.

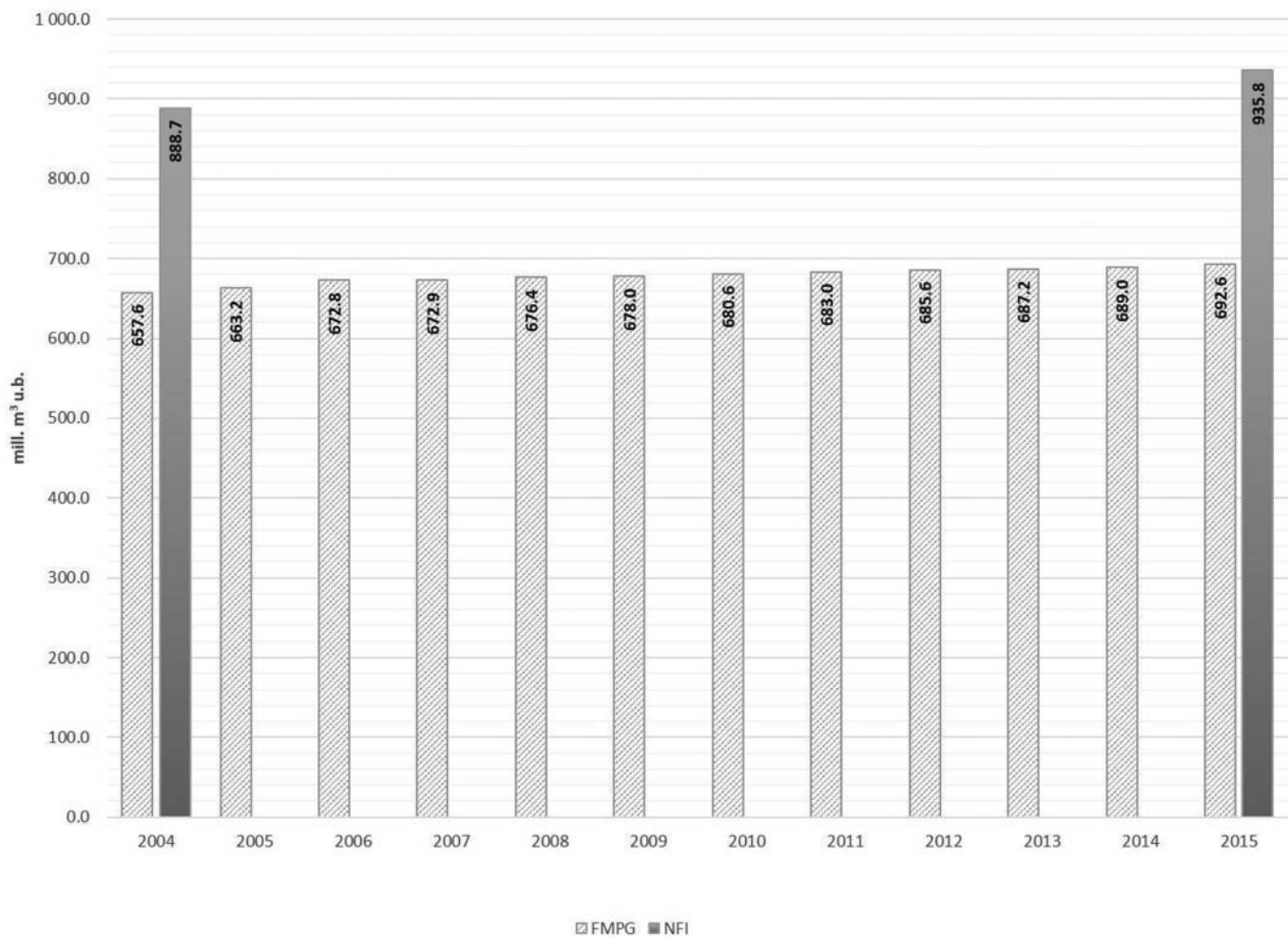
MATERIAL AND METHODS

Traditional data sources for forest policy decision-making in the Czech Republic are Forest Management Plans and Guidelines (FMPG). A national database of FMPGs has been used for evaluation of growing stock, increment and potential wood supply for many years and it is a main source of information for national forestry statistics. Around one tenth of FMPG is updated every year, which means that the data in database are five years old on average. FMPG covers the whole forest area registered in the Cadastre, because it is compulsory for forest owners in the Czech Republic to have Forest Management Plan or simpler form of Forest Management Guideline, which elaboration is fully paid by the Government. It represented 2 647 416 ha in 2005 and 2 668 392 ha in 2015. The growing stock in the FMPG has been derived from volume tables for every stand and tree species during the elaboration of FMPG. The total growing stock from all valid FMPGs is calculated every year as an indicator of sustainable forest management. The increment is also part of the FMPG and it has been calculated on the basis of the Czech growth and yields tables (Černý *et al.* 1996) for every stand and tree species.

The next source of information which is used for forest policy decision-making on national level is removal statistics which is collated by the Czech Statistical Office every year. This statistics is based on a questionnaire survey among forest owners who own more than 200 ha of forest land. Removals realised by smaller forest owners are then estimated according to the results of the survey.

A new data source for forest policy decision-making and for comparison of growing stock, increment, realised felling is the data of the National Forest Inventory (NFI). NFI is based on mathematical-statistical basis. This means that it is not wall-to-wall mapping which is performed by FMPG, but it is based on a field survey which is carried out on sample inventory plots in a defined design. The design is proposed in order to reach required accuracy of the results. The first NFI cycle in the Czech Republic was carried out 2001–2004 and the second one in 2011–2014. The Czech NFI is based on permanent plots and the definition of forest since the second cycle has followed the definition used in FRA and by European National Forest Inventory Network (ENFIN). Compared to the Cadastre and FMPG, small forests are not always taken into account but, on the other hand, for forests growing on other registered land the categories of use were identified. Thanks to this change in the forest definition, in the second cycle it was found out that the forest area is about 9.9% larger compared to the Cadastre and FMPG and it has reached 2, 904, 200 ha (+/- 48, 000 ha). The results of the second cycle for growing stock, increment and fellings were calculated on

FIGURE 1a Total growing stock comparison [million m³]



the basis of repeated measurements of 15,426 permanent plots. A permanent NFI has existed since 2015. NFI has allowed an objective and independent assessment of the actual state and development of forests in the Czech Republic.

Authors have compared growing stock, increment, age class distribution and removals (fellings) statistics from the above defined data sources and have evaluated probable reasons for differences and pros and cons using particular data sources for policy decision-making.

The results were also compared with other countries where NFI has been carried out. National reports collected for the last State of Europe's Forests 2015 publication were used for the evaluation of data sources which are used for international reporting on forest area, growing stock, increment, fellings and removals. These data were collected for the year 2012 and, following national trends, extrapolated to 2015.

RESULTS

Growing stock

The total growing stock is reported annually in national forestry statistics and its calculation is based on the valid

database of the FMPG. According to this data source, the total growing stock has increased every year, but the rate is slowing down, year by year.

In 2005, the NFI1 results indicated that the total growing stock reached 888.7±18.7 million m³ (317.4±3.9 m³/ha), which was more than 35% higher than the official statistics, where the total growing stock based on forest management plans and guidelines made 657.6 million m³ (248.7 m³/ha). In the NFI2, the growing stock definition was revised and updated. The standing dead trees were not taken into account, over bark/under bark coefficients were strictly applied according to Decree No. 84/1996 and statistical methods were updated as well. Despite these facts, the difference was proved by the NFI2 results in 2015 (Adolt *et al.* 2016a), where the total growing stock based on the NFI data rose to 935.8±19.6 million m³ (328.2±4.0 m³/ha), compared to 692.6 million m³ (266 m³/ha) according to the national forestry statistics based on the FMPG (Figures 1a, 1b).

The NFI2 results assessed the total growing stock according to age classes and tree species. These figures are not regularly published as a part of the national forestry statistics based on the forest management plans and guidelines, but it was analysed for comparison with the new NFI2 results.

FIGURE 1b Growing stock per hectare comparison [m³/ha]

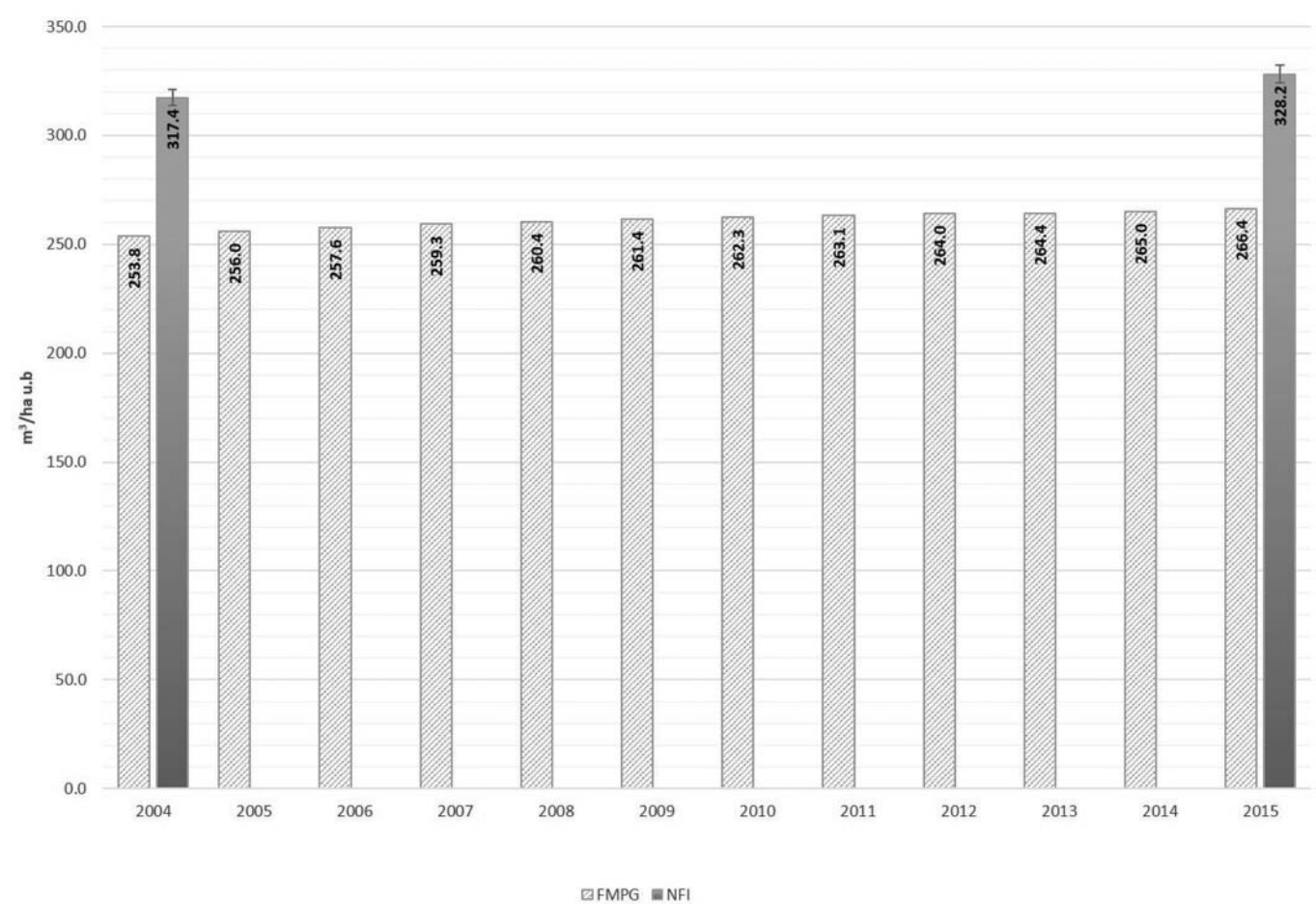


Figure 2a shows the differences in the distribution of growing stock according to age classes expressed as a percentage. This analysis shows low variability compared to the differences in the total growing stock expressed in cubic metres. The biggest difference is in the fourth age class (61–80 years). In NFI1, the sixth age category (51–60 years) had the lowest areal representation and it could influence the result in NFI2, but the difference between the NFI2 and FMPG could have been influenced by salvage felling also, which is mainly caused by the decline of Norwegian spruce stands in the north-eastern part of the Czech Republic. The field survey of the NFI2 was carried out during the years 2011–2015. The survey was able to perform better detection of this salvage felling that increased during last five years and affected spruce stands older than 50 years rather than FMPG which are 5 years old on average.

Figure 2b shows the differences in the distribution of growing stock of timberland per hectare according to age classes expressed in m³/ha.

In Figure 3a the differences in the distribution of the growing stock according to tree species are expressed as a percentage. Based on this analysis we can see that the percentage of growing stock of spruce and pine is higher in the FMPG than in the NFI, but the growing stock of beech, oak and other non-coniferous species is higher in the NFI. This disproportion is more visible when comparing coniferous and non-coniferous

tree species as whole groups. The coniferous tree species account for 81.8% of the total growing stock based on the FMPG, compared to 73.7% based on the NFI. Both data sources confirm a decreasing trend in the growing stock of spruce stands and an increasing trend in the growing stock of non-coniferous tree species.

Figure 3b shows the differences in the distribution of the growing stock of timberland per hectare according to tree species expressed in m³/ha. In this comparison, all tree species have higher average growing stock in NFI compared to FMPG and only spruce shows little decrease.

The potential development trend of the total growing stock was modelled using both of mentioned data sources, however, only a quarter of the final number of NFI2 plots available at the moment (NFI ¼) was used (Adolt *et al.* 2014). In case of the NFI data, the European Forestry Dynamic Model was used and the potential wood supply was modelled using activity probabilities derived from 3,275 pair-wise NFI1-NFI2 plot level observations.

In case of FMPG, the national matrix model based on the Czech forestry legislation was applied. The potential wood supply was modelled using prescribed percentage-shares defined in FMPG. This approach means that all over-matured stands are felled in the first decade with the result that the total growing stock should decrease in the period 2013–2033 but then again it should increase in the following two decades,

FIGURE 2a Total growing stock according to age classes [%]

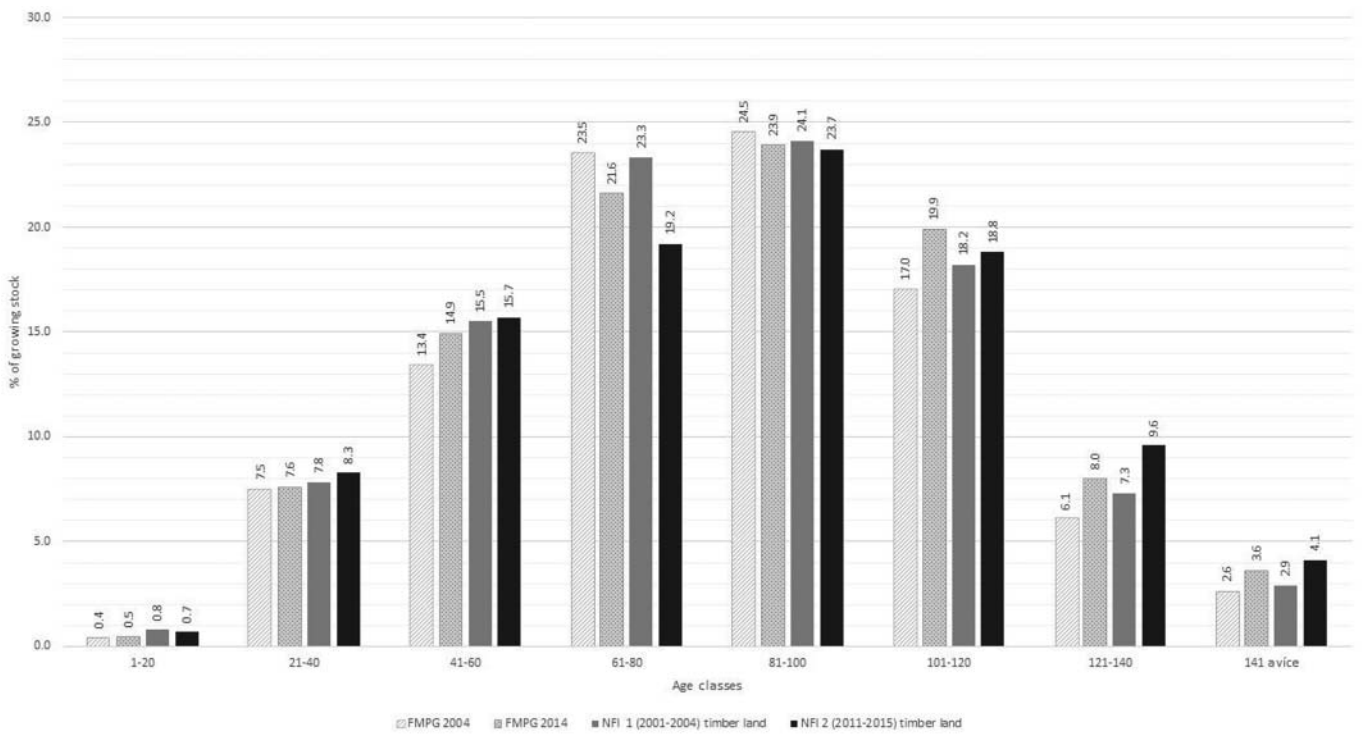


FIGURE 2b Growing stock of timberland per hectare according to age classes [m³/ha]

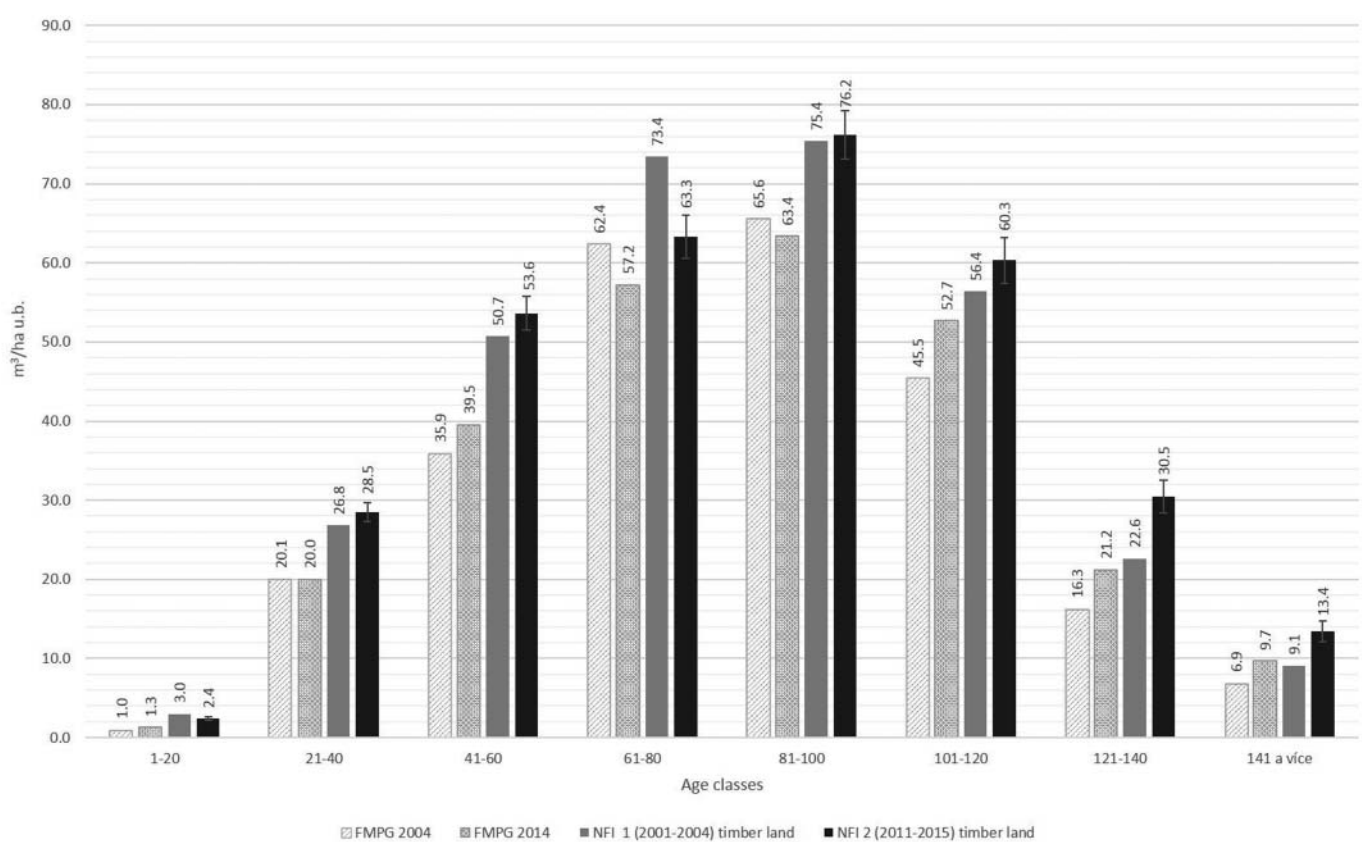


FIGURE 3a Total growing stock according to tree species [%]

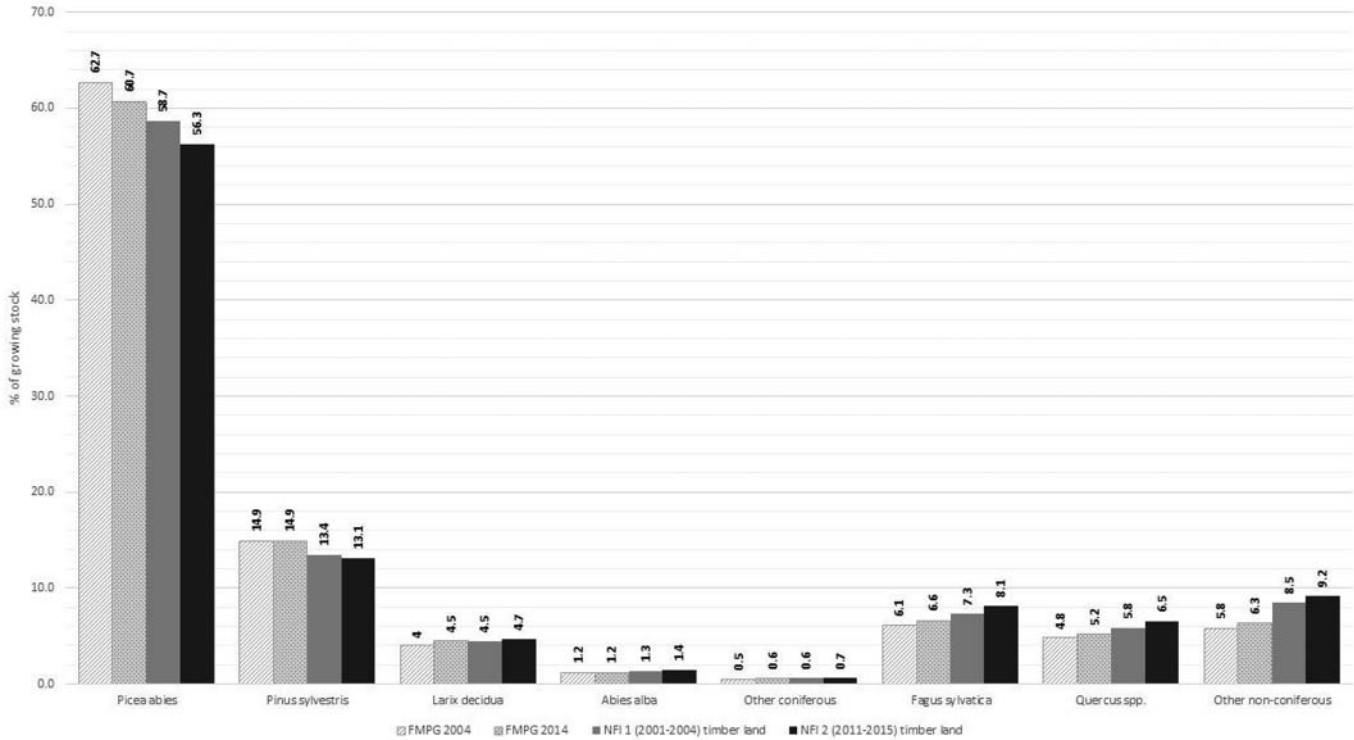


FIGURE 3b Growing stock of timberland per hectare according to tree species [m³/ha]

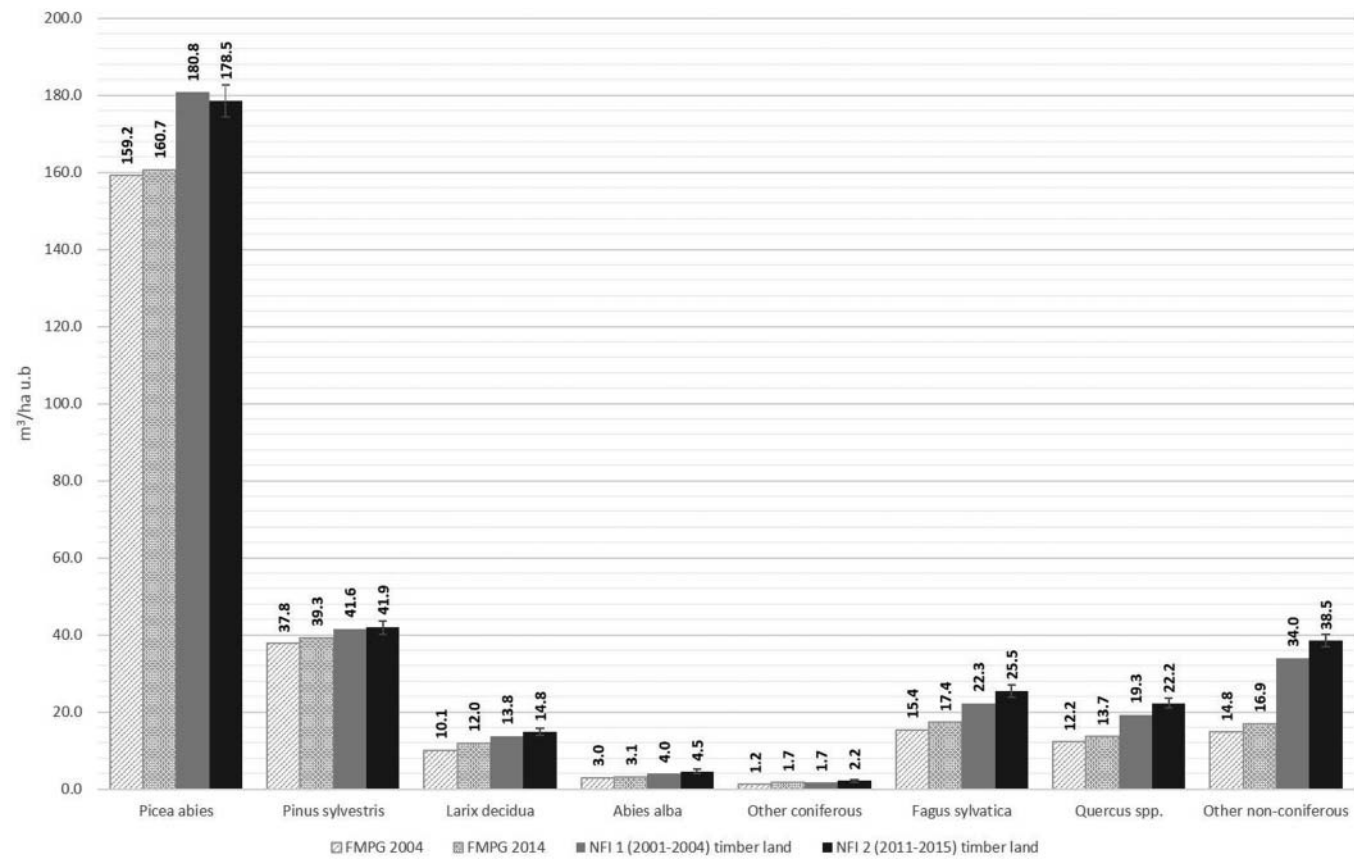
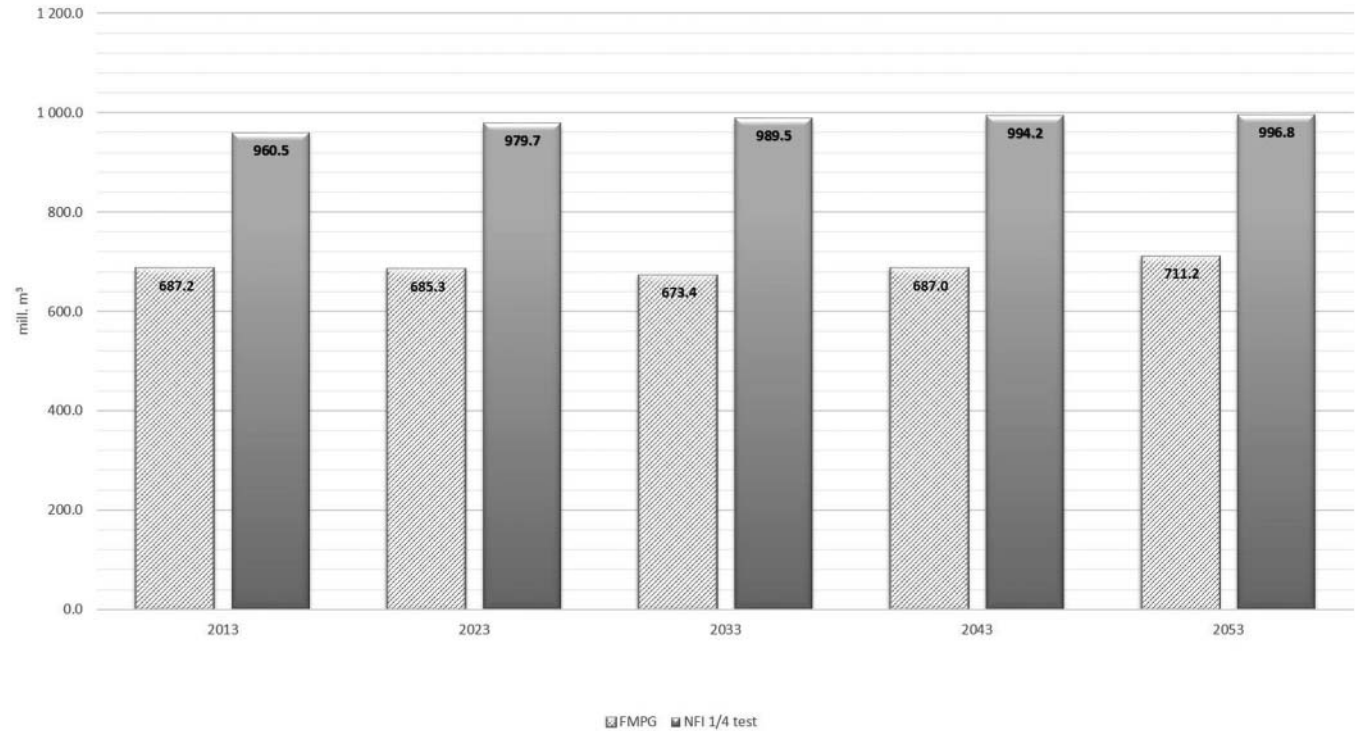


FIGURE 4 Potential development trend of the total growing stock [million m³]



even rise above the level in 2013. Based on the NFI, the total growing stock should increase quite fast in the first two decades, but in the third and fourth decade the rate should slow down significantly.

Increment

The average annual increment accounted for the amounts up to 24.97 ± 0.55 million m³ under bark (u.b.), it means 9.28 ± 0.12 m³ u.b./ha (Adolt *et al.* 2016b) between the NFI1 and NFI2. These figures are higher compared to the official forestry statistics, where the total mean increment, the total current annual increment and the final mean annual increment are calculated based on the FMPG and the Yield tables for the Czech Republic (Černý *et al.* 1996). The average total current annual increment for the period 2004–2014 amounts to 21.13 million m³ u.b., it represents 84.6% compared to the NFI average annual increment.

In Figure 5a, the average annual increment based on the NFI was compared with the total current annual increment from the FMPG database (2004, 2014). The highest difference between both data sources is in the first age class (1–20 years old). A statistically significant difference seems to be in the third age class (41–60 years old) and in the fourth age class (61–80 years old) also, but only when comparing the FMPG 2014 and the NFI. The FMPG 2014 increments distribution as well as the NFI increments distribution copies the current increment curve. The FMGP 2004 increments differ from the current increment curve in the third age class (41–60 years old). This anomaly shifted to the fifth age class (61–80 years old) in 2014, where the increment is lower than in 2004, which is abnormal in the subsequent age classes. This

difference proves the decreasing amount of the growing stock in this age class (Figure 2a).

The highest difference between both data sources in the first age class (1–20 years old) is also confirmed in Figure 5b where the comparison of the increments of timberland per hectare is shown. The differences in other age classes are then lower compared to Figure 5a.

Figures 6a and 6b show the differences in the distribution of the increments according to the main tree species. Based on this analysis, we can see that the increment of spruce, fir and non-coniferous tree species recorded in the NFI is significantly higher than in the FMPG. The difference is also visible between the coniferous and non-coniferous tree species. According to the FMPG, the coniferous tree-species accounted for 79.2% in the total increment in 2004 and for 78.5% in 2014. The NFI results showed the share of the total increment of the coniferous tree species at 75.0%.

The increment based on NFI is influenced by a larger forest area, higher growing stock and, moreover, it is underestimated due to neglecting the increment between the NFI1 measurement and the moment of cutting and mortality.

Age class area distribution

The age categories (a 10-year interval) distribution expressed as a percentage is compared in Figure 7. The age categories development from the FMPG (2004, 2014) was calculated based on the age and timber land area. The main parameters used for the calculation of the age categories distribution from the NFI were tree species, age and a tree crown projection area. If the tree crown projection area was not measured, the dendrometric model was used instead (Kučera *et al.* 2016). As

FIGURE 5a Comparison of the increments according to age classes [million m³ u.b./year]

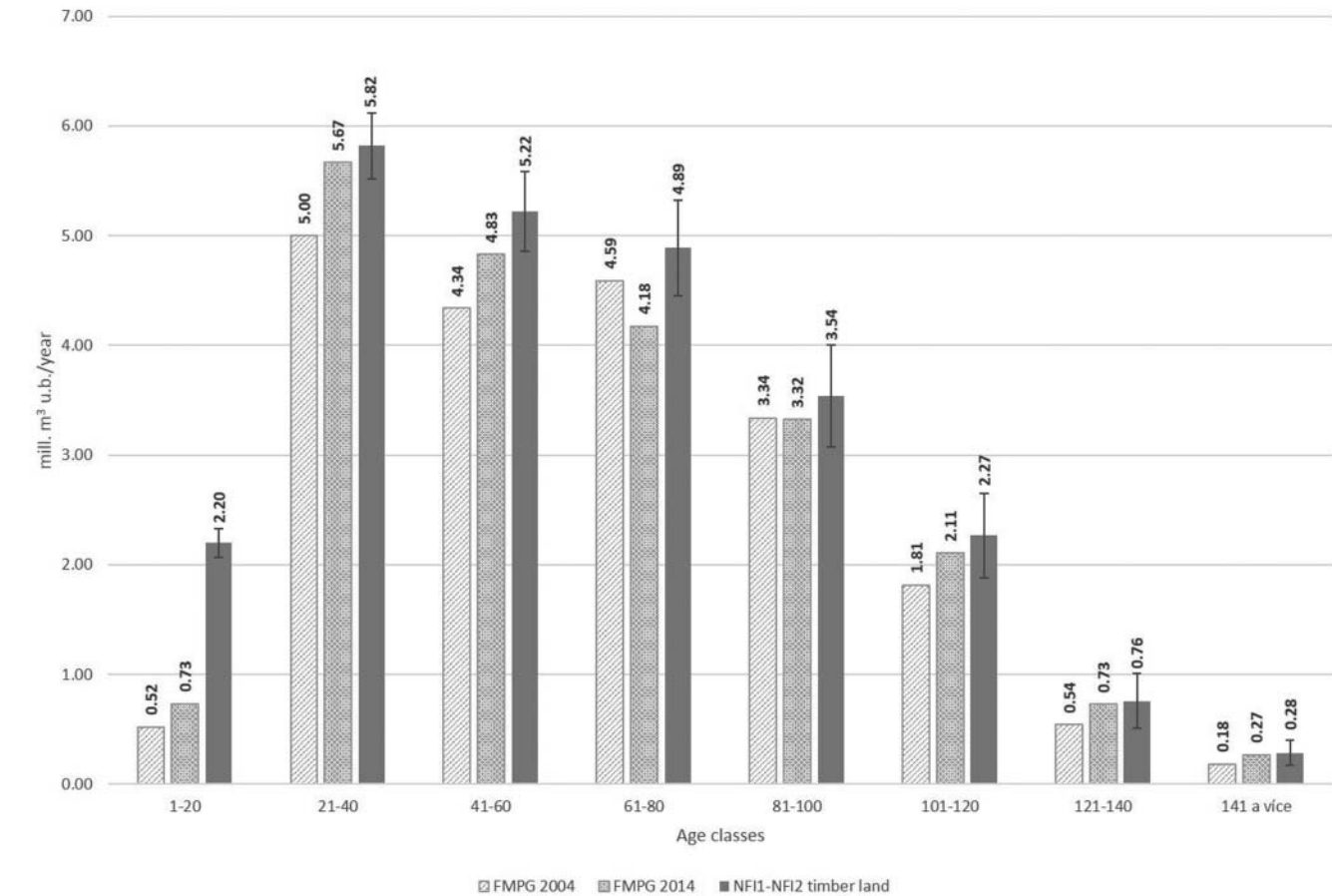


FIGURE 5b Comparison of the increments of timberland per hectare according to age classes [m³ u.b./ha/year]

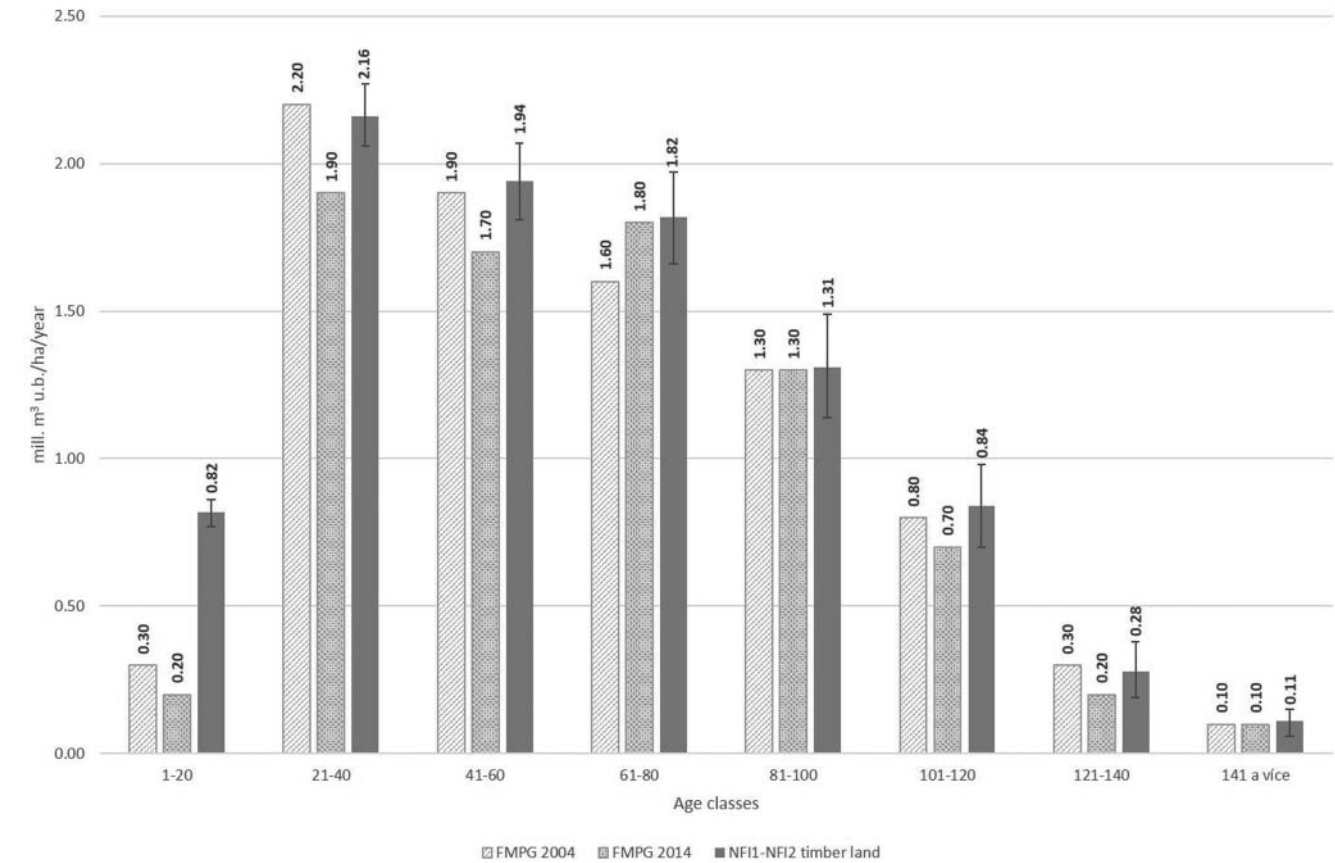
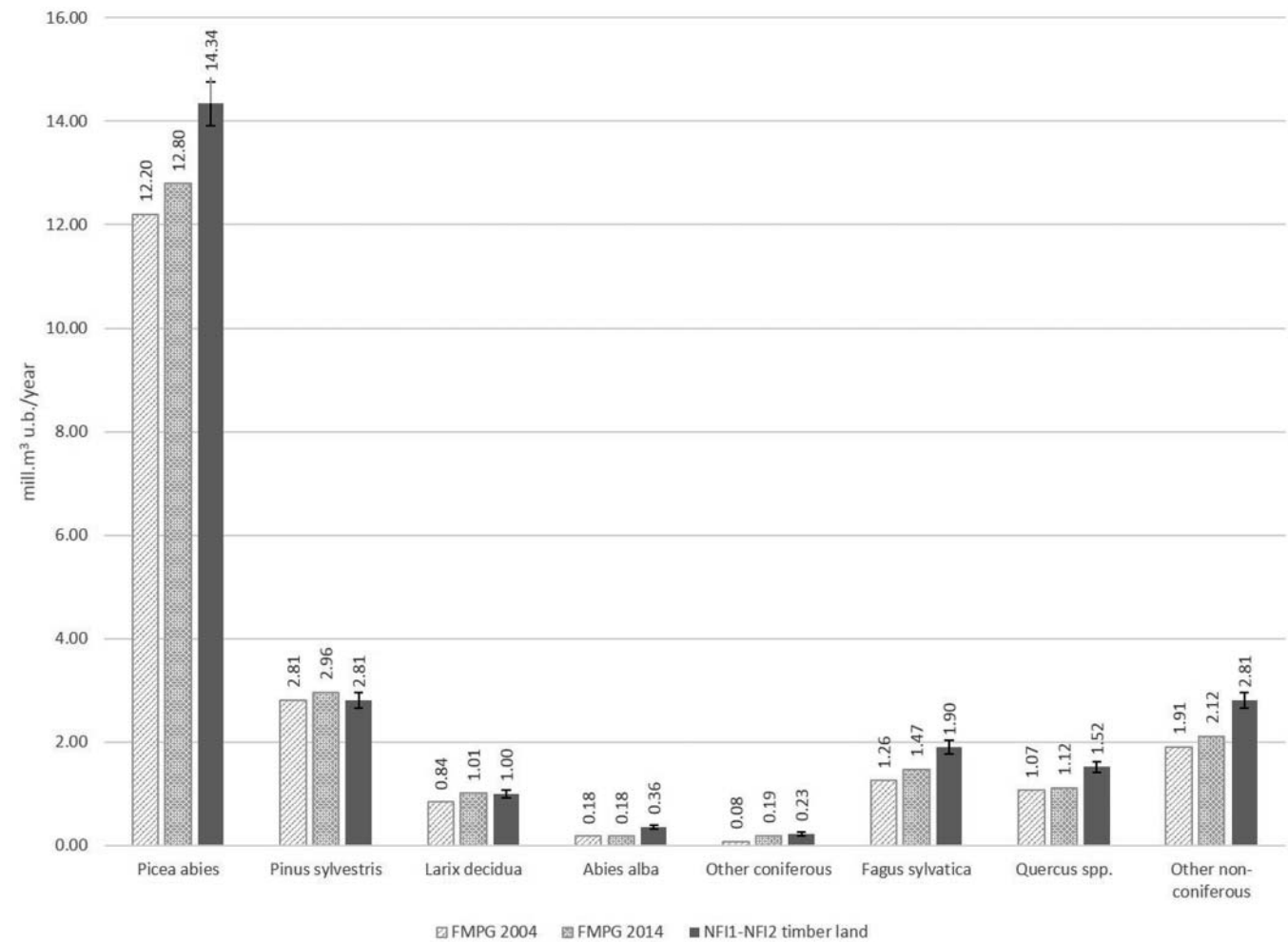


FIGURE 6a Comparison of the increments according to tree species [million m³ u.b./year]



opposed to NFI1 a different methodological approach was used, where subjective canopy cover caused differences in the comparison of age classes' distribution between NFI cycles.

Based on the results, Figure 7 shows that age category 1, age category 2, age category 3 and age category 4 have a significantly higher share according to the NFI results compared to the FMPG. For age categories 5 and 6, the results are comparable while from age category 7 the percentage share of the FMPG has become predominant.

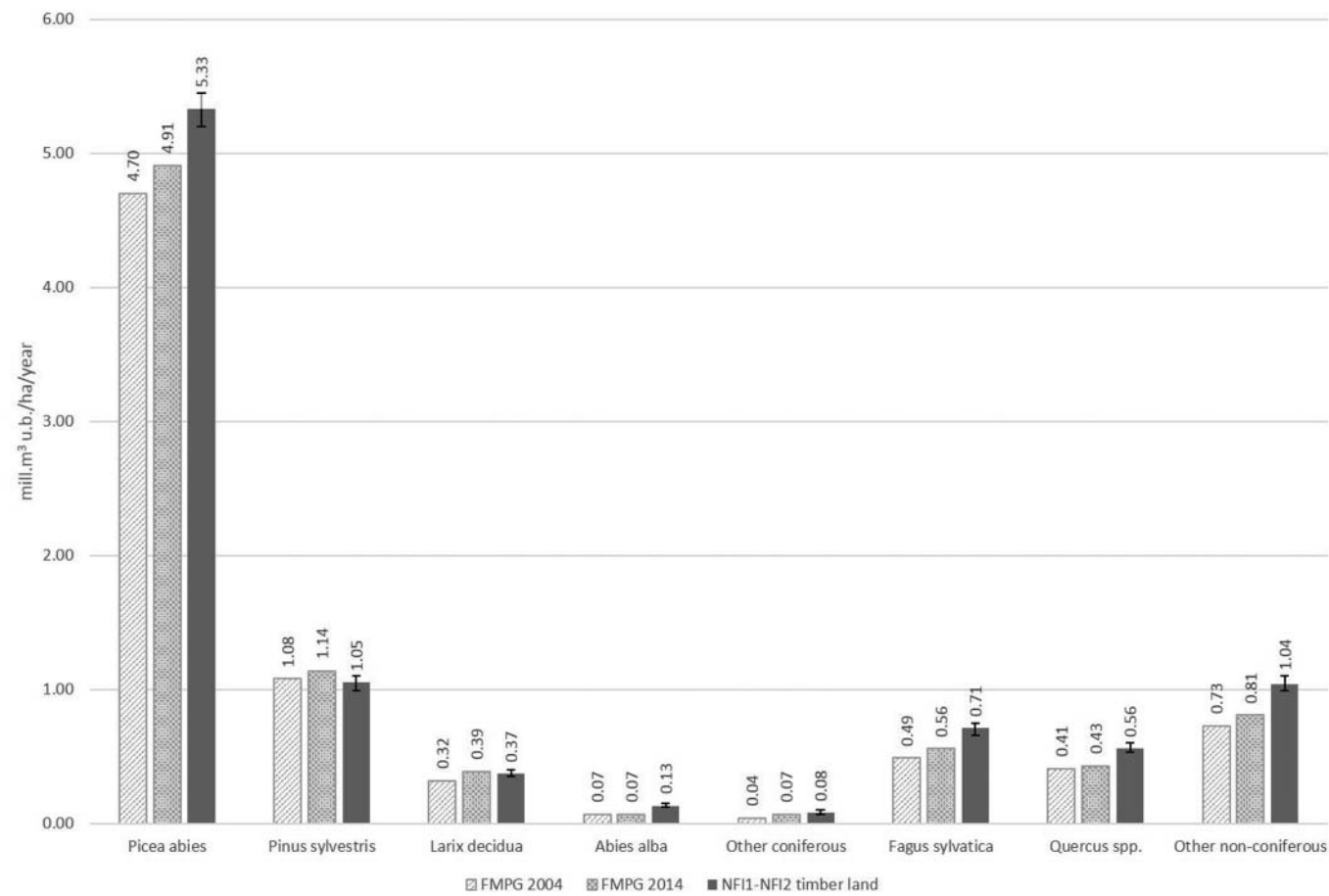
Comparison with the normal distribution of age classes (20-year interval) based on the theory of a normal forest is assessed in Figure 8. In the case of the FMPG 2014, the main disproportions can be seen in age class 2 (21–40 years) and age class 3 (41–60 years), where the difference is about 3% when compared with normality. Regarding age class 2, the difference is caused by a lower share of age category 4 (31–40 years) and regarding age class 3, the difference is caused by a lower share of age category 6 (51–60 years). Age category 7 (61–70 years), is at the same low level of about 7%, but it was balanced by a higher share of age category 8 (71–80 years). This distribution will result in age class 4 (61–80 years) reaching the share of a maximum 13.8% in the next decade, compared to normal distribution of 17.3%.

In case of the NFI2, the main disproportion is in age class 4 (61–80 years), which reached 12.8% when compared to 16.2% in the FMPG only. The difference is caused by a lower representation of age category 7 (61–70 years) which reaches 5.8% only. It confirmed the results from the NFI1, where the lowest share belonged to age category 6 (51–60 years).

Fellings and removals

Fellings are defined in SoEF as “Average standing volume of all trees, living or dead, measured over bark to minimum diameters as defined for “Growing stock” that are felled during the given reference period, including the volume of trees or parts of trees that are not removed from the forest, other wooded land or other felling site. They include silvicultural and pre-commercial thinnings and cleanings left in the forest; and natural losses that are recovered (harvested)”. According to this definition, only NFI repeated measurements can fulfil all required fellings components. The term “removal” differs from “felling” as “it excludes harvesting losses (stemwood) and trees that were felled but not removed and it includes removals from fellings in earlier period and from trees dead due to or damaged by natural causes”. Compared to fellings, removals are reported under bark.

FIGURE 6b Comparison of the increments of timberland per hectare according to tree species [$\text{m}^3 \text{ u.b./ha/year}$]



The NFI2 results, based purely on a statistical approach, brought a new estimation of the total realised felling for the period of 2001–2014. The methodological approach was based on the repeated measurements on permanent plots. Identified standing dead trees from the first cycle which were cut before the repeated survey were not taken into account. Otherwise it is not possible to distinguish the share of trees incorporated in a total amount of fellings which were dead at the moment of their cutting. According to this data source, the mean annual total felling volume for the period of 2001–2014 reached 19.61 ± 0.7 million $\text{m}^3 \text{ u.b.}$, from which coniferous tree species account for 16.78 ± 0.66 million $\text{m}^3 \text{ u.b.}$ (85.5%) and non-coniferous tree species account for 2.84 ± 0.2 million $\text{m}^3 \text{ u.b.}$ (Adolt *et al.* 2016c). The realised felling in the NFI was determined according to a stem status evaluation. The increment between the NFI1 measurement and the moment of cutting was not taken into account, because it is impossible to determine the time when the tree was cut. From this point of view, the estimation of realised felling is underestimated in NFI. This underestimation should not be higher than 10% (Adolt *et al.* 2016c). According to the mentioned methodology, the NFI takes all felled trees between the NFI1 and NFI2 into account, including trees which reached the registration threshold before the NFI2. Not all felled trees are sold on the market or used for one’s own consumption. Based on the

economic or biodiversity requirements, some felled trees are left in the forest and this could be one of the reasons which caused the difference between both results. Such trees should be encompassed as dead wood. Other reasons could be the underestimated FMPG, unreported or underestimated felling by small forest owners.

The official statistics on removals is collected by the Czech Statistical Office which is based on the survey questionnaire among forest owners who possess more than 200 ha of forest land. The amount of felling realised by smaller forest owners is estimated following the results of the survey. There is the potential for the underestimation or overestimation. As the next step, this prediction is compared with the production of trade statistics of the wood-based industry. According to the mentioned methodology, the official removal statistics is mainly focused on the wood realised on the market.

According to this data source, the mean annual total felling volume for the period of 2001–2014 reached 15.74 million $\text{m}^3 \text{ u.b.}$, of which coniferous tree species account for 14.12 million $\text{m}^3 \text{ u.b.}$ (89.7%) and non-coniferous tree species account for 1.67 million $\text{m}^3 \text{ u.b.}$ The record felling of 18.51 million $\text{m}^3 \text{ u.b.}$ was realised in 2007 and was essentially affected by the incidental felling after the gale storm Kyrill.

Official removal statistics represents then 81% of the fellings detected from NFI for the Czech Republic.

FIGURE 7 Comparison of the age categories (10 years) distribution [%]

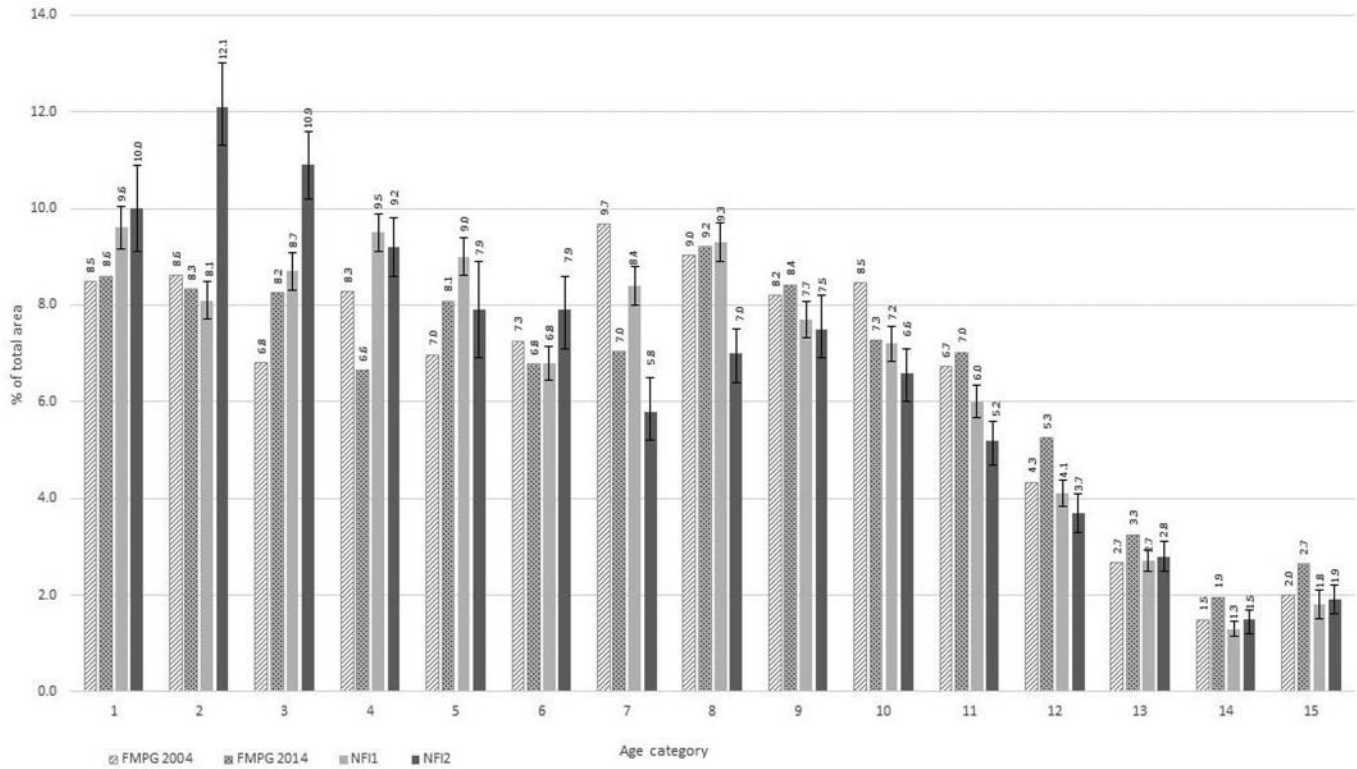
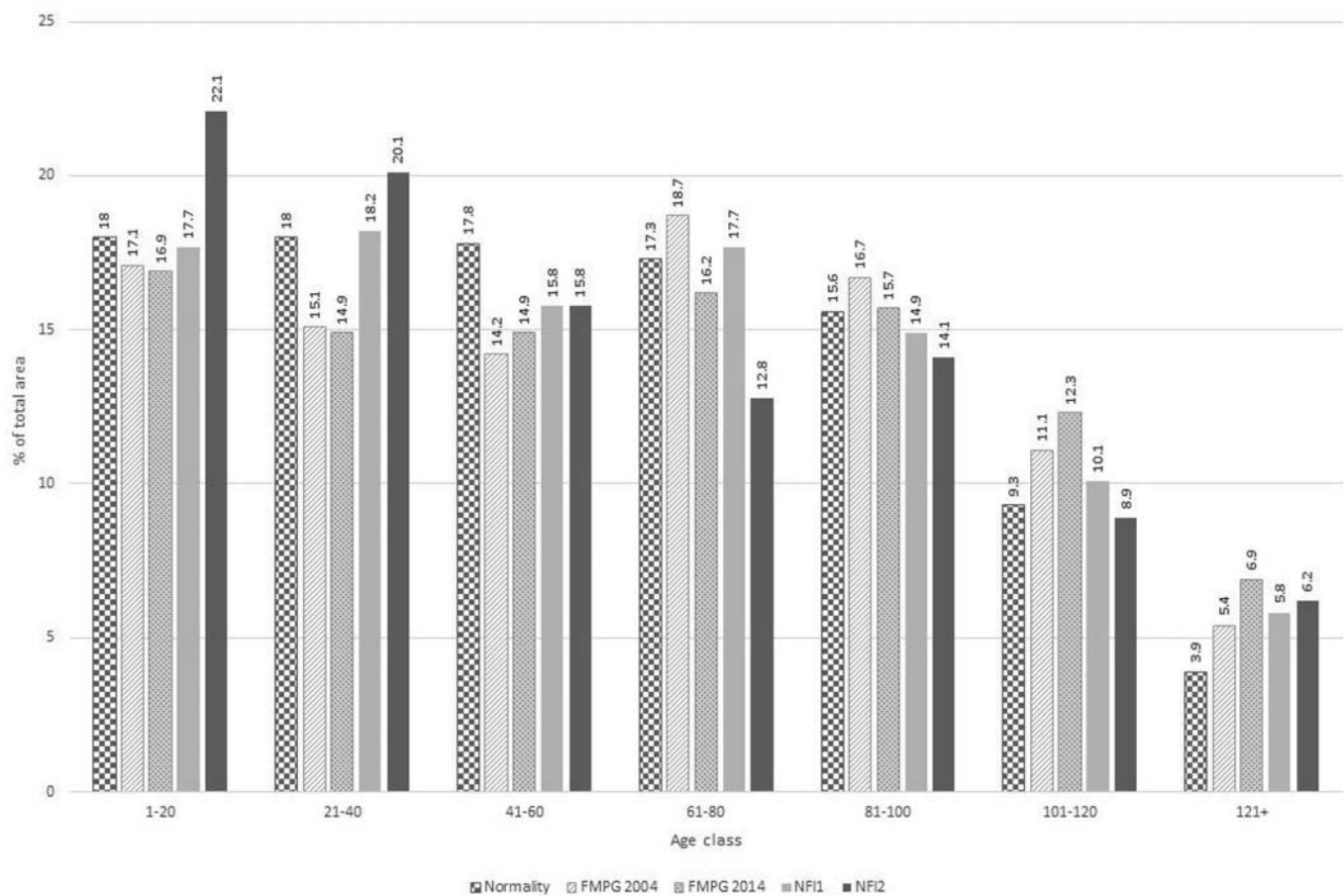


FIGURE 8 Comparison of the age classes (20 years) distribution [%]



DISCUSSION

The main reason for the total growing stock difference in the Czech Republic can be seen in the larger forest area in the NFI, which is about 9.9% larger compared to the Cadastre and FMPG. Another reason is the methodology of the FMPG, where an understorey is not always included in the description of forest stands. Partly, it can also be the underestimation of the growing stock caused by outdated tables used for its determination by the elaboration of the FMPG. In addition, no increment for the period between the elaboration of the FMPG and the current date is reflected (as stated above, the FMPG data is outdated by five years on average).

The underestimation of growing stock derived from forest management plans (stand inventories) by 15–20% and more was also confirmed by Kuliešis *et al.* (2016). Lithuania, Estonia, Denmark, Ireland or Latvia is stated as examples. The main reason is the underestimation of main stand characteristics, as well as a smaller estimation bias.

At an international level the main source of information used by different countries to estimate the growing stock are NFIs, which were used by twenty one countries in SoEF 2015 reporting and as additional sources by other four countries. Stand inventories (FMPG) were used by eight countries while the remaining countries used other sources of information to estimate the growing stock.

In case of the increment, the NFI results shows higher values compared to FMPG, particularly in the first age class. The reason can be seen in the used yield tables (Černý *et al.* 1996) starting from the age of 20. The tables were based on measurements on permanent sample plots with the youngest age class probably excluded from the repeated measurements. Moreover, the yield tables were recommended by authors for use until the year 2010. Therefore they are now outdated and can cause underestimation. The disadvantage of the yield tables is also the fact that they were assembled for homogeneous, even-aged stands of the four main tree species (spruce, pine, beech, oak), yet the stated figures are also used in the prediction of the increment for other tree species. In the NFI, the increment can be measured on all plots and trees with repeated measurements. The average annual increment is thus determined regardless of tree species composition, stocking or the community status of trees.

Kuliešis *et al.* (2016) mentioned that data on gross annual increment (GAI) including its components estimated from stand inventories are low reliability and are characterized by frequent systematic errors compared to NFIs. GAI assessed from stand inventories is 20–30% lower compared to NFIs due to a lower growing stock estimate and inaccurate stand characteristics.

As regards the GAI estimation, the main sources of information used by most countries in Europe are NFIs which were used by nineteen countries. Four countries have calculated the GAI as a percentage share of the growing stock and other eight countries which were able to report on increment have used other sources of information, e.g. yield tables or

tree rings measuring. Twelve countries did not report on GAI in SoEF 2015 reporting.

Regarding the age class distribution, the results from both data sources indicate a disproportion in the representation of age class 4 (61–80 years) and particularly age class 3 (41–60 years). This will probably have a negative impact on the potential wood supply in the next 20 years.

In case of felling and removals, the difference between the official statistics and the NFI results was also proved in Switzerland where, according to official statistics, the average annual felling reached 86% of the estimation based on the NFI (Brändli *et al.* 2010). The underestimation of the official removal statistics was also ascertained in Germany, where removal statistics represents 75% of the fellings detected from NFI. The comparison of the side-used calculated data for felling with the supply-based data from the official statistics shows that there was a difference of about 16.1 million m³ in 2013 (Jochem *et al.* 2015). Also Italy, according to the SoEF questionnaire, used a conversion factor based on NFI results and official removal statistics for fellings estimation for reporting in SoEF 2015.

Main reasons for differences between fellings (over bark) and removals (under bark) in the Czech Republic seem to be private forest owners' self-consumption, especially fuelwood self-consumption. Private forest owners in European countries have different grades of freedom in their decision on how much wood they are allowed to harvest from their properties without approval. This amount of fellings is legal, but in many cases not registered in removal statistics. In the Czech Republic, forest owners without an approved forest management plan or without a protocol concerning the acceptance of adopted forest management guidelines, a simplified form of forest management plans for small forest owners, are allowed to cut 3 m³/ha/year without state forest administrative permission. Forest area covered by forest management guidelines has reached around 378,000 ha, which means means that around 1.13 million m³ u.b. can be legally felled every year without state forest administrative permission. It represents about 7% of annual volume of removals according to official removal statistics.

The amount of bark presents another difference between fellings and removals. Bark is not taken into account by taking delivery in sawmills and it is subsequently used for energy production or as a product for mulching.

Other differences can arise from using different methods of trees measurement. These include trim allowance, rounding of lengths and diameters according to national rules, purchasing wood by weight or measurements by electronic taking delivery in sawmills. Mantau *et al.* (2016) assumed the difference between standing tree volumes and available volumes of felled trees that reach the mill site around 10%.

The difference also consists in wood left in forests, either for economic reasons or for biodiversity requirements. A reliable source of information on wood left in the forests is NFI, provided deadwood occurrence and its measurement are incorporated in the field survey. In case of the Czech Republic lying merchantable deadwood longer than one meter comprises 24.1 million m³ o.b., it is 8.6 m³ o.b./ha.

CONCLUSIONS

The aim of the article was to compare the current state and potential development of the main indicators of sustainable forest management, in particular the growing stock, the increment, and the realised fellings, based on two different data sources which are available in the Czech Republic, and subsequently analysed their pros and cons in terms of their suitability to support policy decision-making on different levels.

Regarding the growing stock, the NFI2 results confirmed that the growing stock in the FMPG is significantly underestimated and reached 74% of the total growing stock estimated by NFI only and 79% compared to the growing stock on the timber land declared by the cadastre. According to the above mentioned results, the more reliable data source for the estimation of the total growing stock on both the national and regional level is the NFI data. On the lower level and stand level, the FMPG can be used for the determination and prediction of the growing stock only.

There is a slightly less significant difference in the case of the increment determination compared to the growing stock. The average total current annual increment from the FMPG for the period of 2004–2014 represents 84.6% compared to the NFI average annual mean increment. The main difference between both data sources was recorded in the first age class (1–20 years) where the increment from the FMPG only reached 33% of the NFI increment. The difference is caused by the use of the yield tables (Černý *et al.* 1996) which are outdated, assembled for homogenous, even-aged stands of the four main tree species and so they can cause the underestimation. The NFI data is thus more reliable and complex for both the growing stock and for the estimation of the increment at the national and regional level. In the case of a permanent NFI measurement, it is possible to detect dynamic changes in the increment development in shorter intervals. The static yield tables can be used for the lower level and the stand level.

Regarding the realised felling, quite a large difference was recorded among official removal statistics performed by the Czech Statistical Office and the fellings based on NFI results. According to the official statistics, the average annual felling reached 81% of the estimation based on the NFI. The official statistics is mainly focused on the wood realised on the market. This statistics could be partly improved or verified in the future, using new satellite images and LIDAR data from the satellite Sentinels for detecting felled areas. The NFI model and results will definitely be improved by incorporating the estimation of the increment on the felled trees.

Both data sources have an important information value for policy decision-making at both the state and regional level. The NFI data is a more reliable data source for modelling the potential wood supply at the national and regional level for the future, because they are based purely on a statistical approach, they take all forests into account according to the NFI definition, are elaborated upon by one methodology and by one independent subject. A felling probability is more over based on real forest owners' behaviour during the last years and not on the theoretically prescribed fellings percentage from the forest legislation.

The FMPG can mainly be used for an ad hoc analysis of the potential wood supply on a local level, e.g. the potential loss of wood supply from an intended protected area, the potential wood supply for an intended local sawmill, etc. The FMPG has remained the main tool for ensuring sustainable forest management for forest owners as well as for state forest administration. They have also played an important role in policy decision-making, e.g. by planning and evaluating subsidies paid from national and international sources or by applying different intentions with forests through declared forest categories.

On the other hand, the FMPG is not able to determine other important indicators such as the occurrence and volume of dead wood, realised felling, damage inflicted by game, forest regeneration or changes in soil conditions. An important role of the NFI should be to serve as an independent source of information on forests and, in the case of a permanent NFI, to provide complete, up-to-date results every five years. In the case of the need to collect some new information on forests, a permanent NFI is a very flexible tool for the incorporation of new requirements in measurement in comparison to the FMPG where the collected information is being cut down.

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